SCRIPT-CONTROLLED ACL

Restrict Record Access Based on Field Value

Project Content / Journey Map

SCENARIO

A user requests access to a record in an application. The system evaluates a field value on that record and uses a script-controlled Access Control List (ACL) to decide whether the user can view or access the record.

ENTER

The user signs in and attempts to open a protected record. The application identifies the requested record, the logged-in user, and the relevant field value.

STEP	INTERACTION / EXPERIENCE	THINGS / TOUCHPOINTS	PLACE	PEOPLE	POSITIVE MOMENTS	NEGATIVE MOMENTS	AREAS OF OPPORTUNITY
1. Open protected record	User selects a record or opens its record URL.	Application UI / record page	Application portal or system	Logged-in user	Clear indication that access is being checked.	Unexpected access denial or unclear permission message.	Show a clear access status and explain what information is required.
2. Identify field value	The system reads the field used by the ACL condition.	Record fields / database	Backend record service	Application / ACL script	Correct field value is available for evaluation.	Missing, empty, or unexpected field values can cause access problems.	Validate the field and define a safe default for missing values.
3. Execute ACL script	The ACL script compares the record field value with the user's role, identity, or permitted value.	Script-controlled ACL	Authorization layer	ACL script / user context	Fast and consistent authorization decision.	Incorrect script logic may expose or block records.	Keep the script simple, documented, and testable.
4. Allow or deny access	The system grants access when the rule is satisfied; otherwise it denies access.	Access-control response	Application / security layer	User and system	User gets the correct record access.	Denied users may not understand why access was blocked.	Return a suitable permission message without revealing sensitive data.
5. Audit and review	The system records the authorization result for troubleshooting and review.	Logs / audit records	Administration or monitoring area	Administrator / security team	Easy investigation of access decisions.	Insufficient logs make errors difficult to diagnose.	Log relevant events while avoiding unnecessary sensitive information.

GOALS & MOTIVATIONS

- Help me access records that I am authorized to use.
- Help me avoid seeing records that are outside my permitted field value.
- Help administrators control record visibility using a clear rule.
- Help the application apply the same access logic consistently.
- Help administrators troubleshoot denied or unexpected access.

POSITIVE MOMENTS

- Authorized users can access the correct records.
- Access decisions are applied automatically.
- Field-based rules can support different access requirements.
- A script can centralize the authorization logic.

NEGATIVE MOMENTS

- Users may receive an access-denied message unexpectedly.
- Incorrect field values can lead to an incorrect access decision.

- Complex ACL scripts can be difficult to maintain.
- Poorly defined rules can create inconsistent access behavior.

AREAS OF OPPORTUNITY

- Validate field values before making the authorization decision.
- Document the ACL condition and script logic.
- Test allowed, denied, empty, and unexpected field-value cases.
- Provide a clear but security-conscious access-denied message.
- Maintain audit information for authorization troubleshooting.

EXTEND

After deployment, administrators can review access logs, update the field-based rule when requirements change, and test the ACL whenever roles, field values, or record structures are modified.

CORE PROJECT FLOW

User Request → Read Field Value → Run ACL Script → Compare Rule → Allow / Deny Access → Audit Result

EXPECTED OUTPUT

The expected output is a secure record-access mechanism in which the ACL script checks a specified field value and permits or restricts access according to the defined rule. Unauthorized users should not be able to view protected record data.

Template basis: The uploaded reference uses a journey-map structure with scenario, steps, interactions, touchpoints, positive/negative moments, goals, opportunities, and extension.